

Sthuthi Poojary

Aspiring Machine Learning and AI Engineer

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Summary

Final year AIML student with strong foundations in Data Structures, Machine Learning, Neural Networks and Computer Vision. Experienced in building real-time applications, ML systems, and SaaS analytics pipelines using modern web technologies. Passionate about open-source development and scalable software systems.

Technical Skills

- **Languages:** C, C++, Python, JavaScript, SQL, R
- **ML & AI:** PyTorch, TensorFlow, Gradio, SHAP,
- **Backend and Databases:** Firebase, PostgreSQL, MySQL
- **Tools:** Git, Docker, Power BI, Notion
- **Core:** Data Structures & Algorithms, Machine Learning, OOP, DBMS, System Design

Projects

Federated Learning for Skin Cancer Detection.

2026

- Designed and implemented a privacy-preserving Federated Learning system (FedProx + Weighted FedAvg) for 7-class skin cancer classification on HAM10000 across 3 simulated hospital clients, achieving 88% accuracy and 0.97 macro AUC on a dual-backbone EfficientNet-B4 + MobileNetV3 fusion model.
- Roadmapping edge deployment on NVIDIA Jetson Nano via TensorRT INT8 quantisation and knowledge distillation.

Multi AI Agent System

2026

- Designed and built a 5-agent system (Orchestrator, Researcher, Coder, Writer, Critic) from scratch with no orchestration framework dependencies.
- Implemented ReAct agent loop with tool calling (web search, Python REPL, calculator), rolling memory buffer, and ChromaDB long-term memory
- Built a Critic loop with structured JSON scoring that retries failing outputs until quality threshold ($\geq 7/10$) is met.

PCOS Early Detection System

2025

- Built a dual-model ML pipeline (XGBoost + SHAP) for early PCOS detection, achieving 98.4% ROC-AUC and 100% recall using clinical features, with a symptom-only model reaching 94.3% ROC-AUC for lab-free screening, obtained through model evaluation.
- Implemented online learning via exponential moving average across user check-ins, full explainability through SHAP feature contributions and data visualization through React.js dashboard.

Experience

Core Team Member - Sahyadri Open Source Community.

2025 – Present

- Organized and hosted multiple technical workshops and community events.
- Actively contributed to open-source projects.

Smart India Hackathon (SIH)

2024, 2025

- Shortlisted among the top 10 teams in internal SIH rounds for two consecutive years.

GirlScript Summer of Code

2025

- Contributed to open-source projects with 5+ merged pull requests.

Education

B.Tech in CSE(Artificial Intelligence and Machine Learning) (CGPA: 9.63)

2023 – Present

Sahyadri College of Engineering and Management

PCMB (Percentage: 95.8%)

2021 – 2023

St. Aloysius Pre-University College

Achievements & Activities

- Organizer at SOSC – conducted workshops and coding events.
- Mentor at HackHarbour 3.0 – guided 100+ students.
- Hosted O(n)slaught – inter-college competitive programming contest.
- Solved more than 250+ questions in Leetcode to improve problem solving.